DATA 608 / KNOWLEDGE & VISUAL ANALYTICS

How much do we get paid?

The salary story changes with the job title — and the state.

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THE QUESTION

Same field, different ways to get paid

A descriptive comparison of annual mean wages—not an individual salary prediction.

06

OCCUPATIONS

Official BLS SOC categories; not exact matches for every industry job title.

16

SELECTED STATES

Six-role comparisons across a deliberately selected subset of U.S. states.

49

JURISDICTIONS

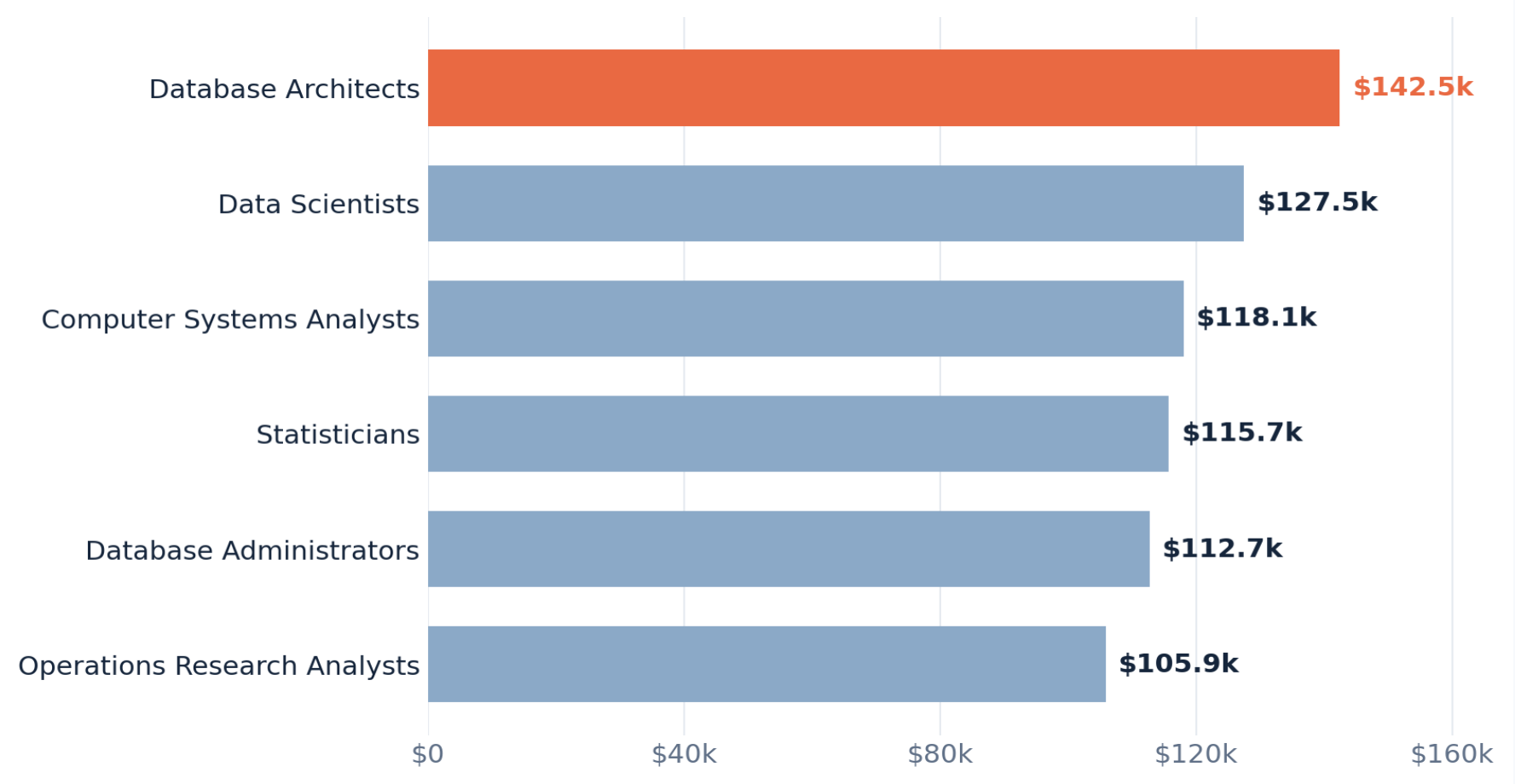
Published Data Scientist wage observations in the source table.

Important: equal-state averages are not U.S. national means; nominal wages are not cost-of-living adjusted.

FINDING 01 · ROLE

Database Architects lead the six-role comparison

Equal-weight average of the 16 selected state mean wages, annual USD



$142.5k

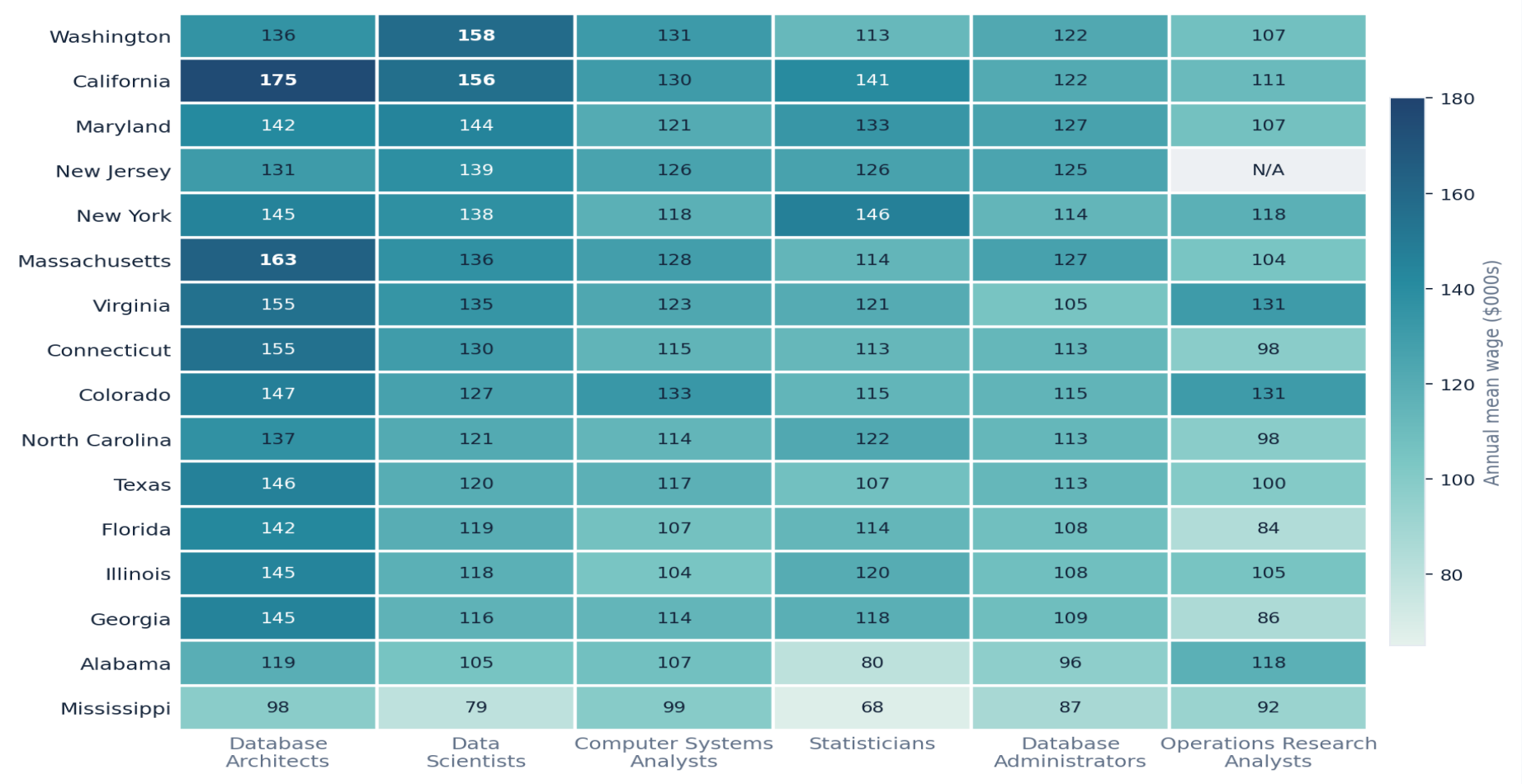
Database Architects average $14.9k more than Data Scientists in this selected-state comparison.

These are averages of state means—not a worker-weighted U.S. estimate.

FINDING 02 · ROLE × STATE

There is no single salary ranking for every state

Darker cells indicate higher annual mean wages ($000s); gray means unavailable



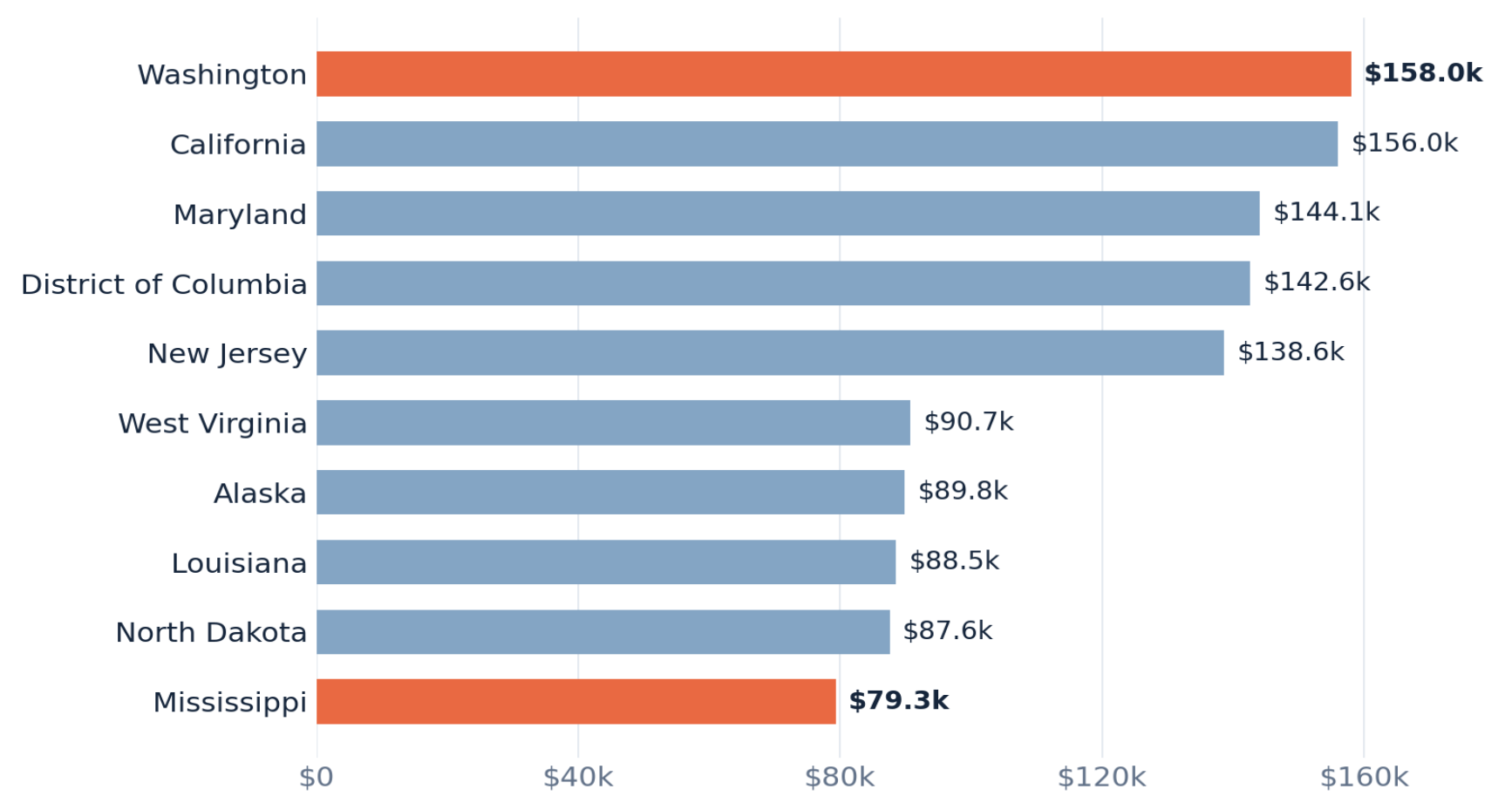
Look across & down

Compare roles within a state (across a row), then compare states within a role (down a column).

California Database Architects: $175k; Mississippi Statisticians: about $68k.

Data Scientist wages nearly double between extremes

Five highest and five lowest published jurisdictions, annual mean wages



$78,750 gap

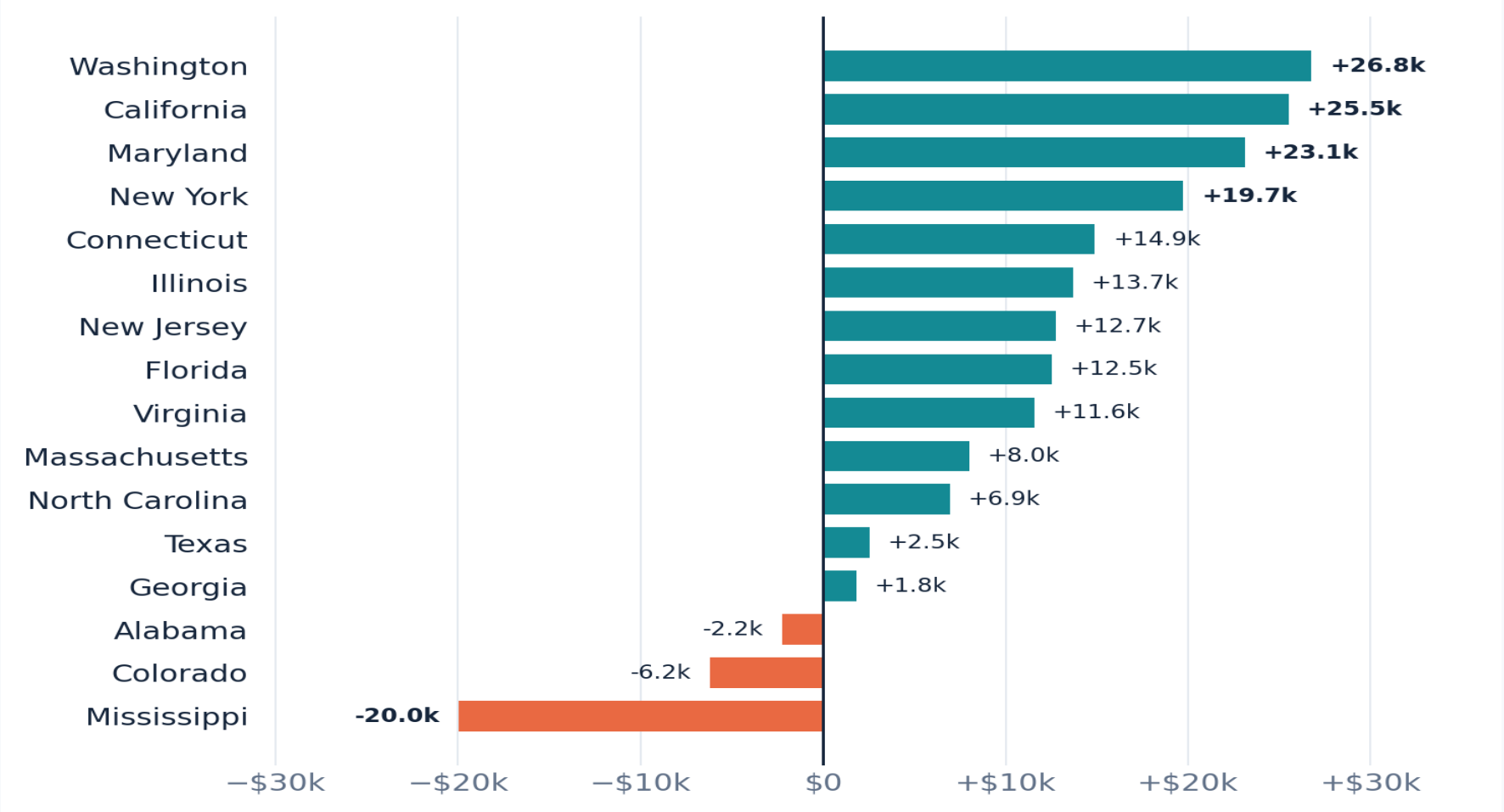
Washington: $158,040
Mississippi: $79,290

The higher figure is about 1.99× the lower—not a measure of after-tax purchasing power.

FINDING 04 · INTERACTION

The Data Scientist pay advantage is not universal

Data Scientist mean wage minus Computer Systems Analyst mean wage, 16 selected states



The sign flips

Washington: +$26.8k

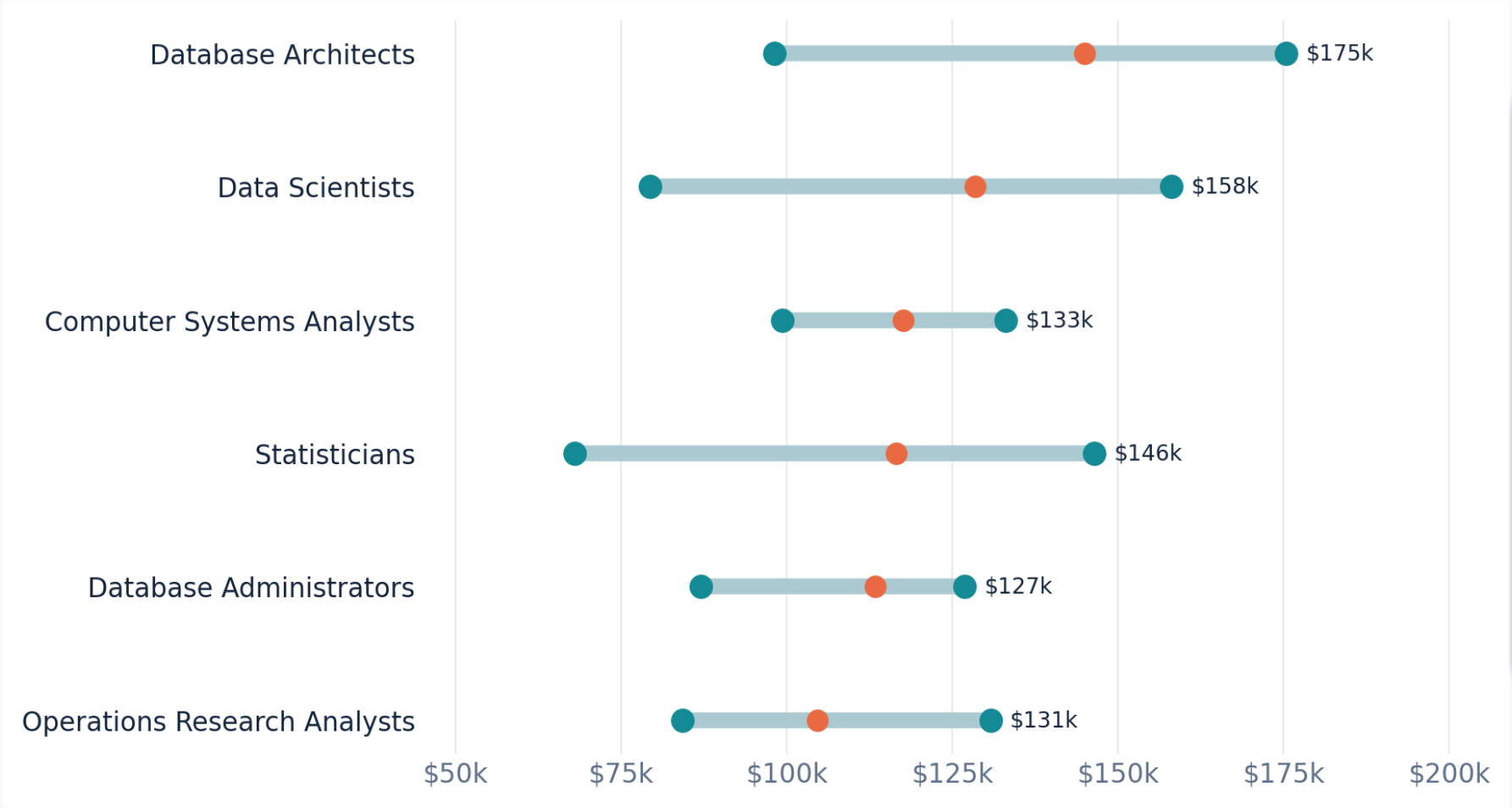
Mississippi: −$20.0k

A positive difference favors Data Scientists. A negative difference favors Systems Analysts.

FINDING 05 · DISPERSION

A role can have a wide wage range across states

Endpoints show lowest and highest selected-state mean; orange dot = median state wage



Spread matters

The role ranking hides geographic variation.

A range plot reveals how far apart state mean wages can be within each occupation.

This uses the same selected 16 states.

INTERPRETATION

What these comparisons can - and cannot - tell us

Read every salary figure in the context of its definition and coverage.

WAGE MEASURE

Annual mean employee wage, not median, bonuses, benefits, or total pay.

OCCUPATION MAPPING

Data Engineer / Data Analyst / Business Analyst are not one-to-one SOC categories.

GEOGRAPHIC SCOPE

Six-role matrix covers 16 selected states; 49 entries for Data Scientists only.

CAUSALITY & VALUE

These patterns do not explain why wages differ or compare purchasing power.

CONCLUSION

How much do we get paid? It depends on both role and location.

Three key findings reveal how occupation and geography influence annual mean wages

01

ROLE

Database Architects top the selected-state role ranking at about $142.5k.

02

STATE

Data Scientist mean wage spans $79.3k to $158.0k across published jurisdictions.

03

COMBINATION

The Data Scientist vs. Systems Analyst pay gap switches sign in three of 16 states.

Bottom line: a single headline salary masks meaningful differences in both occupations and geography.

SOURCES & METHODS

Transparent data provenance and reproducible code

Salary values: third-party BLS-attributed tables, not direct official BLS bulk downloads

DATA & OCCUPATIONAL DEFINITIONS

BLS Occupational Employment and Wage Statistics (OEWS), May 2025.

Six detailed SOC occupations: 15-2051, 15-1243, 15-1242, 15-1211, 15-2031, and 15-2041.

Wage entries transcribed from SignalBleed BLS-attributed occupation tables.

ANALYSIS & LIMITATIONS

Python notebook and local CSVs support all calculations.

Equal-state averaging; missing wages retained as missing.

16 states for joint role/state analysis, 49 jurisdictions for Data Scientists.

No geographic price adjustment or causal inference.

Official BLS tables: bls.gov/oes/tables.htm • Source mirror: signalbleed.com/jobs/